SIGNSIGHT: EMPOWERING ACCESSIBILITY WITH TAMIL SIGN LANGUAGE RECOGNITION USING MACHINE LEARNING

25-26J-404

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

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August 2025

TAMIL SIGN LANGUAGE (TSL) SMART LEARNING & EDUCATION PLATFORM

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date: 25/9/19

Signature of the co-supervisor: Date: 25/09/19

Abstract

This proposal presents a web based Tamil Sign Language Learning & Education Platform designed for learners in Sri Lanka. The platform brings together four parts: a structured course that teaches both static and dynamic signs, a real-time practice lab that uses the webcam to give instant and easy to follow feedback on each sign, a friendly AI learning companion that can chat in Tamil (text or speech) and demonstrate signs in everyday contexts, and a teacher and parent dashboard that shows progress and learning patterns to guide support.

The teaching approach is straightforward: short lessons, frequent practice, and clear corrections that build confidence. The system adapts to each learner's pace and preferred style (games, stories, or step by step drills) and gently adjusts when the learner seems bored or frustrated. Privacy and safety are built in from the start: video stays on the device by default, only learning results are stored, and guardians control data for children.

Overall, the platform aims to make TSL learning practical, motivating, and inclusive for anyone, anywhere. learners can practice every day, communicate more clearly, and involve family and teachers in a positive way.

Keywords – ***Tamil Sign Language, Intelligent Tutoring Systems, Gesture Recognition, Adaptive Learning, Emotion-Aware Feedback***

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LIST OF ABBREVIATIONS

Abbreviations	Description
ASR	Automatic Speech Recognition
TSL	Tamil Sign Language
AI	Artificial Intelligent
ITS	Intelligent Tutoring Systems
AWS	Amazon Web Services
PWA	Progressive Web App
CDN	Content Delivery Network
SUS	System Usability Scale
NLP	Natural Language Processing
CNN	Convolutional Neural Network
ML	Machine Learning
BKT	Bayesian Knowledge Tracing

1 INTRODUCTION

Tamil Sign Language is a critical medium of communication for Tamil speaking Deaf and hard of hearing communities worldwide. The global need is substantial: more than five percent of the world's population lives with hearing loss, and tens of millions are children underscoring the importance of accessible, high quality sign language learning at scale[1]. Yet most learning tools still emphasize passive content (videos, dictionaries, flashcards) over the kinds of guided practice that actually change performance during production. Recent work in sign learning systems explicitly identifies a gap in real time practice and evaluation, noting that many applications teach vocabulary but do not help learners correct their signing while they practice [2].

Communication support is also often one directional (e.g., speech to sign or sign to text), which limits everyday usefulness. Research in assistive communication argues for bidirectional, multimodal pipelines that combine speech-to-sign with live gesture recognition to enable natural two way interaction and practice in context [7]. Building on that direction, our component focuses on learning (not just translation), so learners can rehearse TSL actively, receive immediate corrections, and use conversational scenarios to connect vocabulary to real life.

Concurrently, advances in AI directly match what independent learners need. A recent systematic review shows that AI most effectively supports self directed learning through personalized tutoring, query/response assistance, timely feedback and recommendations, and interactive dialogue a set of functions repeatedly linked to improved outcomes[3]. Within this, conversational agents are highlighted as especially useful for sustaining interaction and knowledge construction during self study[4]. Complementary ITS literature emphasizes adaptive assessment, real time feedback, and data backed educator insights while cautioning about ethics, transparency, and scalability, which any modern platform must address from the start [5].

This proposal introduces a web based TSL Learning & Education Platform designed for global access on ordinary devices. It unifies: a curriculum guided course for static and dynamic signs; a real time practice lab that uses a webcam to analyze handshape, movement, location, and orientation and then delivers clear, actionable feedback; a friendly AI learning companion that chats in Tamil (text or speech) and runs role play scenarios to encourage contextual signing; and an insights dashboard for teachers and parents that surfaces progress, common error patterns, and targeted recommendations. A browser first design is chosen for scalability, low cost, and inclusive reach principles supported by prior web platform research that stresses individualized service, interaction, and sustained development of networked learning communities[6].

Pedagogically, the platform favors short lessons, frequent practice, and simple corrections that build confidence. It adapts to a learner's pace and preferred mode (games, stories, step by step drills), and it can gently adjust when the learner appears bored or frustrated (switching activity types, slowing the pace, offering hints). In parallel, the conversational layer follows evidence from multimodal assistive research linking Tamil text/speech to TSL practice so that learning moves beyond isolated signs to functional, two way communication[7]. Finally, because emotional nuance is central to meaningful signing, the design considers emotion and context, reflecting findings that purely linguistic pipelines struggle with affect and cultural variation; our approach adds emotion aware practice to keep learning natural and motivating [8].

Finally, this component responds to documented gaps with an end to end platform that is practice centric, dialog driven, adaptive, and explainable. It leverages AI functions known to benefit self learners (personalized tutoring, feedback, recommendations, dialogue) while meeting ITS expectations for transparency and educator oversight. The result is a practical path to scalable, privacy conscious, and inclusive TSL learning for anyone, anywhere so learners can practice every day, communicate more clearly, and include families and teachers in a positive feedback loop [3][5].

1.1 Background and Literature Survey

Sign language education platforms have historically emphasized fixed lessons and drill based exercises, with limited support for real time practice and evaluation. The Master-ASL project explicitly identifies this gap and positions learning and assessment together to address it (noting WHO prevalence and "considerable deficiency in real time practice and evaluation")[1] Its mobile app blends multimedia instruction and game-oriented assessments useful signals for engagement, but still representative of first generation, single learner experiences without deep personalization or dialogue[1].

Across broader learning science, a 2025 IEEE Access systematic review synthesizes how AI best supports self directed learning (SDL): the dominant functions are personalized, on demand tutoring, query or response support, timely feedback and recommendations, and interactive dialogue through conversational agents[2]. The same review details that real-time feedback/recommendations consistently improve SDL outcomes and satisfaction, while a lack of prompt feedback is a barrier to progress[2]. It also documents the value of conversational agents for high interactivity learning and knowledge consolidation in self regulated contexts . More broadly, the review concludes that AI tools increase access, reduce cognitive load, and enhance motivation when learners pursue autonomy driven goals key prerequisites in SDL models[2].

Within ITS, recent surveys chart the evolution from rule based tutors to ML/NLP-driven systems and now generative AI tutors capable of adaptive, human like dialogue and fine grained learner modeling. These systems aim to monitor learner behavior and dynamically adjust pedagogical strategies, yielding better engagement and retention while introducing challenges of transparency, bias, and deployment at scale[3].

From a platform perspective, classic web-learning work stresses interactive services, whole process support, and individualized delivery in networked environments and learning communities[4]. It also recommends user customized displays and integration patterns that reduce navigation friction principles we adopt in our web application[4]. Complementary work on digital learning platforms further enumerates environments every modern system should enable: interactive learning, Q&A, personalization, evaluation & incentives, statistical feedback, and rich resources with explicit guidance to design around learners' abilities, habits, and needs[6].

For sign technology pipelines, contemporary speech and sign research reports progress by combining NLP with CNN/ML for gesture recognition and translation, but also flags persistent challenges: dialectal variation, expressive nuance, and real-time constraints that can degrade user experience if not addressed. These findings motivate our emphasis on visual, camera only interaction (no wearables) with robust, low latency feedback and culturally adaptable content.

Takeaway for this component. Evidence converges on four design imperatives for a next gen TSL platform:

(1) Continuous, real-time practice with instant feedback
(2) An AI tutor/companion for dialogue, Q&A, and motivation
(3) Personalization via learner modeling and recommendations
(4) A web first, interactive platform that supports individualized delivery and learning communities.

These imperatives are directly aligned with the reviews above and address the "practice and evaluation" gap observed in current sign learning apps.[1][2]

1.2 Research Gap

(1) **Limited real time assessment and corrective feedback for signing.** Many apps teach static signs and quizzes; few deliver low latency, frame by frame corrective feedback during continuous signing (the "real-time practice and evaluation" deficiency)[1].

(2) **Under used conversational pedagogy in SDL contexts.** Review evidence shows query response and interactive dialogue are powerful for self learning, yet are sparsely applied in mainstream platforms (only one study explicitly targeted learners' Q&A support; conversational agents remain under deployed)[2].

(3) **Personalization and learner modeling not first class in sign learning tools.** ITS scholarship emphasizes adaptive strategies and data driven support, but sign specific learning systems rarely integrate mature learner models, mastery tracking, or style aware delivery[3].

(4) **Expressivity and variation.** Speech and sign systems still struggle with dialects and emotional nuance; most classroom apps do not scaffold emotive/role play practice that reflects how sign language is actually used[5].

(5) **Platform integration.** Prior web platforms highlight individualized service, Q&A, and continuous support, but sign learning tools seldom unify these affordances into one coherent, global, web first experience[4][6].

Features	Research Paper [1]	Research Paper [2]	Research Paper [3]	Research Paper [4]	TSL Learning Platform
Conversational practice (Tamil text/speech ↔ TSL role-play)	**No.** Lesson/quiz oriented.	**Strongly recommended.** Interactive dialogue aids SDL.	**Supported.** Dialogue for tutoring	**Supported.** Interaction & Q&A patterns.	✓
Personalization & learner modeling	**Light.** Progress/assessments.	**Core.** Personalized tutoring, recommendations.	**Core.** Adaptive pathways, mastery modeling.	**Supported.** Individualized services.	✓
Explainable feedback (why it's wrong)	**Minimal.** Scores/grades.	**Recommended.** Feedback clarity supports SDL.	**Critical.** Transparency for educator trust.	N/A	✓
Teacher/parent dashboards (actionable insights)	**Basic.** Learner results.	**Supports.** Recommendations & progress.	**Supported.** Data-backed educator insight.	**Supported.** Statistical feedback in web platforms.	✓
Web-first, no special hardware; offline support	**Yes.** Mobile/web delivery at scale.	N/A	N/A	**Yes.** Web interactivity & personalization.	✓
Learning impact beyond model metrics	**Limited.** App-level results.	**Encourages.** Outcomes & satisfaction.	**Encourages.** Learning gain studies.	Platform-level.	✓

Table 1.1 Comparison of the existing research methods and proposed method

1.3 Research Problem

Despite the development of sign language recognition and learning technologies, the learners of TSL still experience education and communication barriers. Mobile and web-based platforms like Master-ASL [1] have shown that sign lessons with quizzes can take place through mobile and web-based apps, but most of them cover the American Sign Language and offer partial support to Tamil. These systems are based on the recognition and evaluation of the static nature and hence the gap in providing the corrective feedback in real-time in dynamic and natural signing.

Studies on AI-based education indicate that the most effective features are individualized tutoring, in-person communication, and feedback in a timely manner[2][3]. These features are not commonly exercised by existing sign-learning applications, however. Students are used to training without being adjusted to their speed or mood, and there is little opportunity to achieve engagement-conscious coaching and an individual lesson planning, both important on motivation and long-term improvement.

Multi-modal translations approaches, including the Tamil speech-to-sign pipeline by Kowsigan[5], show that it is possible to perform multimodal translations, but not the pedagogy. They have difficulties with dialectal differentiation, expressiveness, and responsiveness in the moment. Traditional web learning platforms[4] are personalized and interactive yet fail to answer the needs of the TSL education. This results in an imperative research gap: the absence of a scalable, browser based TSL platform that combines real time coaching, conversational role-play, personal adaptation and explainable teacher parent insights to ensure learning is effective and inclusive.

2 OBJECTIVES

2.1 Main Objective

1. Develop a comprehensive learning platform for Tamil Sign Language (TSL):

- Create an interactive system for teaching TSL with structured lessons, assessments, and quizzes.

- Enable users to learn both static and dynamic signs, improving communication for the hearing-impaired community in Sri Lanka.

2. Enhance dynamic sign recognition using machine learning:

- Build a system capable of recognizing and interpreting dynamic TSL gestures using advanced computer vision techniques.

- Implement real-time recognition of gestures, improving accessibility and communication for TSL users.

3. Bridge the gap between spoken Tamil and TSL through audio/video conversion:

- Develop a tool that converts Tamil audio and video inputs into corresponding TSL signs.

- Provide an intuitive solution that helps hearing-impaired individuals understand spoken language via visual representation.

4. Develop emotion and context-aware TSL enhancement for children:

- Design interactive exercises to help children, including those with cochlear implants, recognize and express emotions through TSL.

- Track progress and provide real-time feedback to improve expressive and communication skills in children.

5. Foster inclusivity and accessibility for the hearing-impaired community in Sri Lanka:

- Create a user-friendly platform that enhances communication, education, and interaction opportunities for the hearing-impaired.

- Ensure that technology promotes social inclusion, making daily life and learning easier for TSL users.

2.2 Specific Objective

- Develop a systematic TSL program to be taught to both the static signs and the dynamic signs, broken down into explicit lessons, so that learners can slowly build up on their vocabulary and communication.
- Create motion-rich demos and interactive tasks for letters, words, and sentences, with slow-mo, loop, and frame-by-frame.
- The system reads quiz results to build weekly personalized lesson packs, and Mentor Chat sends short tips and micro-lessons to bring learners back on track.
- Develop an AI Learning Companion that works with text and speech in Tamil where learners can practice by using it in role play conversations in real world contexts.
- Individualize the learning process by monitoring every learners learning and determining areas of difficulty and automatically differentiating lessons and activities to suit their speed and learning style.
- Incorporate feeling sensitive adjustment that will identify when learners are bored, frustrated or busy and then shift the speed or the type of activity to keep learners motivated and interested.
- Give a Teacher / Parent Dashboard showing the progress of learners, any hard indicators and give simple weekly suggestions to instruct follow up and support further.

- Make the platform accessible and secure through the provision of global accessibility, safety by providing it as a browser based platform with offline backup, inbuilt privacy safeguards, parental controls on children, and a simple interface to use by anybody of any age.

3 METHODOLOGY

This section explains the methodology followed in designing and developing the TSL Learning & Education Platform. The approach combines AI-based sign recognition, personalized tutoring, conversational practice, and analytics dashboards, ensuring the system is interactive, scalable, and globally accessible. The project uses an iterative design and development model, where prototypes are tested and refined based on usability feedback, accuracy benchmarks, and learning outcomes.

3.1 System Description

Overview

The platform is a Tamil-first web app (PWA) that teaches Tamil Sign Language through a personalized step-by-step track (letters → small words → simple sentences). Learners study with motion-rich demos and screen-based interactive practice (MCQ, match, order-frames, finger-spelling, gloss-fill). The system builds weekly personalized lesson packs, applies recovery patterns when someone falls behind, and lets mentors send micro-lessons. It works offline, stores only progress data, and requires no learner webcam.

Users & Roles

- Learner: studies lessons, completes tasks, earns checkpoints/badges, receives weekly packs.
- Mentor/Teacher: views dashboards/alerts, sends micro-lessons, curates content.
- Admin/Author: adds signs/lessons, uploads demo videos/animations, creates tasks, publishes modules.

3.2 Data Collection

The project requires a dataset of Tamil Sign Language videos. Data will be collected from multiple signers of different ages, genders, and backgrounds to ensure diversity. Each sign will be recorded in multiple conditions (lighting, background) for robustness. Signs will be annotated with phonological features (handshape, location, movement, orientation, non-manual markers).

- Static Dataset: Core vocabulary (nouns, verbs, alphabets, numbers).
- Dynamic Dataset: Daily phrases, sentences, and role-play conversations.
- Ethics: Data will only be collected with informed consent. For children, guardian consent and anonymity safeguards will be applied. Raw video will remain confidential, and only derived keypoints will be stored.

3.3 Tools & Tchnologies

- Frontend: Next.js for building the web interface.
- Backend: Node.js 20 microservices (practice, personalization, mentor), PostgreSQL 16, object storage for media.
- Personalization**:** BKT/DKT + spaced repetition → weekly personalized lesson packages.
- Mentor Chat**:** Rasa / LLM API to send micro-lessons and tips.

3.4 Implementation & Deployment

Implementation will follow Agile sprints:

Phase 1 – Core Lesson Delivery
Implement lesson player with motion-rich demo playback, initial task types (MCQ, Match), and basic authoring tool.

Phase 2 – Extended Task Engine and Progress Tracking
Add order-frames, finger-spelling, and gloss-fill tasks. Introduce checkpoints, badges, and offline lesson packs.

Phase 3 – Personalization and Recovery Mechanisms
Integrate BKT with spaced repetition. Generate weekly personalized packs and apply recovery patterns (review-refresh, skill-focus, pace-slow, spaced-revisit).

Phase 4 – Mentor Integration and Dashboards
Develop Mentor Chat for micro-lessons and tips. Provide dashboards for learner progress and alerts.

Phase 5 – Pilot Testing and Refinement
Conduct pilot with schools, NGOs, and home learners. Collect feedback, measure usability and performance, and refine system.

Deployment Strategy:

Deploy as a PWA with React 18 and Next.js 14 frontend. Backend with Node.js microservices, PostgreSQL, Object Storage, and Redis. Use cloud deployment with containerization, CI/CD pipeline, and secure media delivery through CDN.

Evaluation Targets:

- Learning gain: +20% pre to post.
- Usability: SUS ≥ 75, ≥90% completion.
- Operations: publish sign ≤5 min, video start ≤2 s.
- Engagement: ≥60% weekly return, recovery within 1 week.

3.5 Evaluation

Evaluation will be conducted at three levels:

1. Model Evaluation
   - Static signs: top-1/top-5 accuracy, per-class F1.
   - Dynamic sequences: sequence-level F1, gloss edit distance.
   - Latency profiling across mid-range devices.
2. Learning Effectiveness
   - Pre-test vs. post-test learner accuracy and vocabulary size.
   - Drop-off rate comparison: adaptive vs. fixed learning path.
   - Usability testing (System Usability Scale, SUS score ≥80).

3. User Feedback
    - Qualitative interviews with learners, teachers, and parents.
    - Engagement metrics (practice time, retries, hint usage).
    - Teacher/parent dashboard usefulness ratings.

3.6 System Architecture

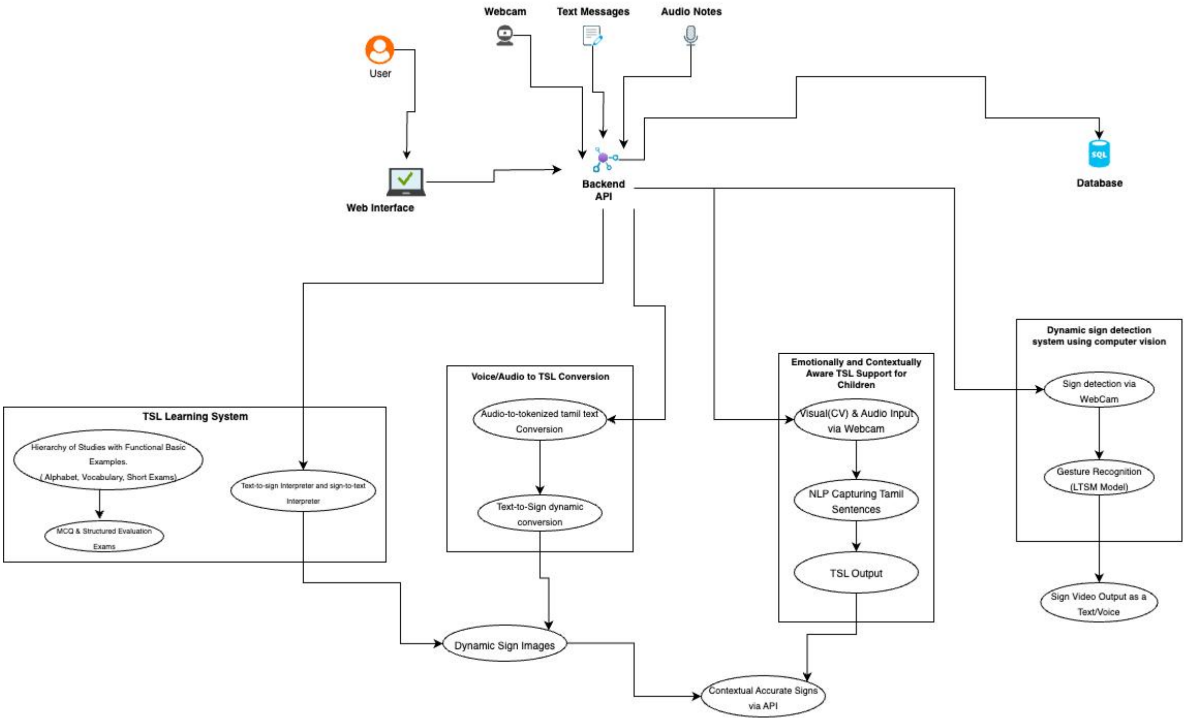

Figure 3.1 System Architecture

3.7 Component Overview Diagram.

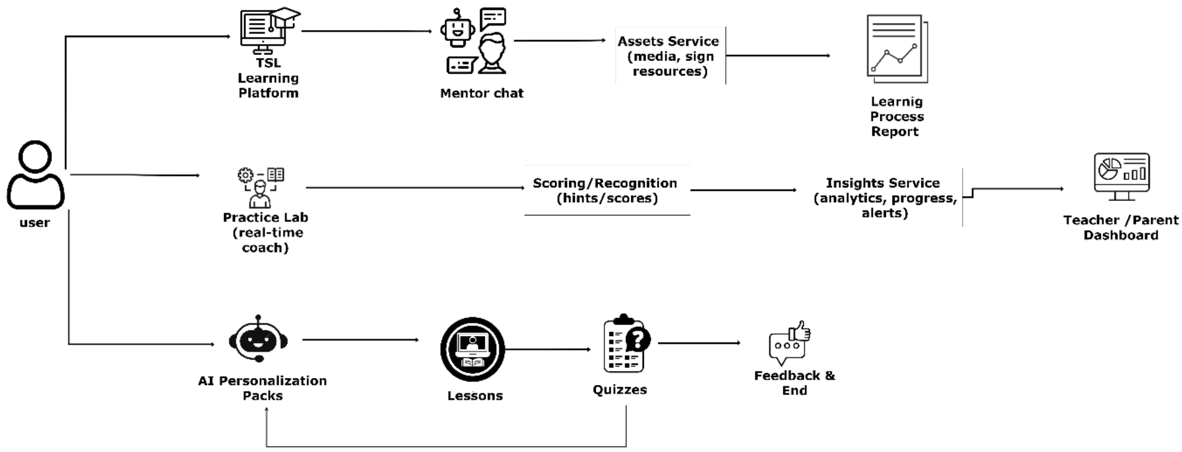

Figure 3.2 Component Overview Diagram

End-to-End Flow :-

1. Learner opens the browser app, picks a lesson, watches a demo, and does screen-based tasks.
2. The Practice Service marks answers, returns hints, and logs each attempt.
3. The Personalization Service updates per-skill mastery (BKT), chooses next items, and composes the weekly personalized pack. If the learner is out-of-track (low scores, repeated errors, inactivity), it assigns a recovery pack (review-refresh / skill-focus / pace-slow / spaced-revisit).
4. The Insights Service turns activity into simple charts and alerts (e.g., “movement topic is weak”) for the learner and mentor dashboards.
5. The Assets Service delivers secure media links (videos/animations) and supports offline packs so lessons run with poor internet.
6. Mentor Chat can push a short micro-lesson and tips based on alerts.

3.8 Work Breakdown Structure

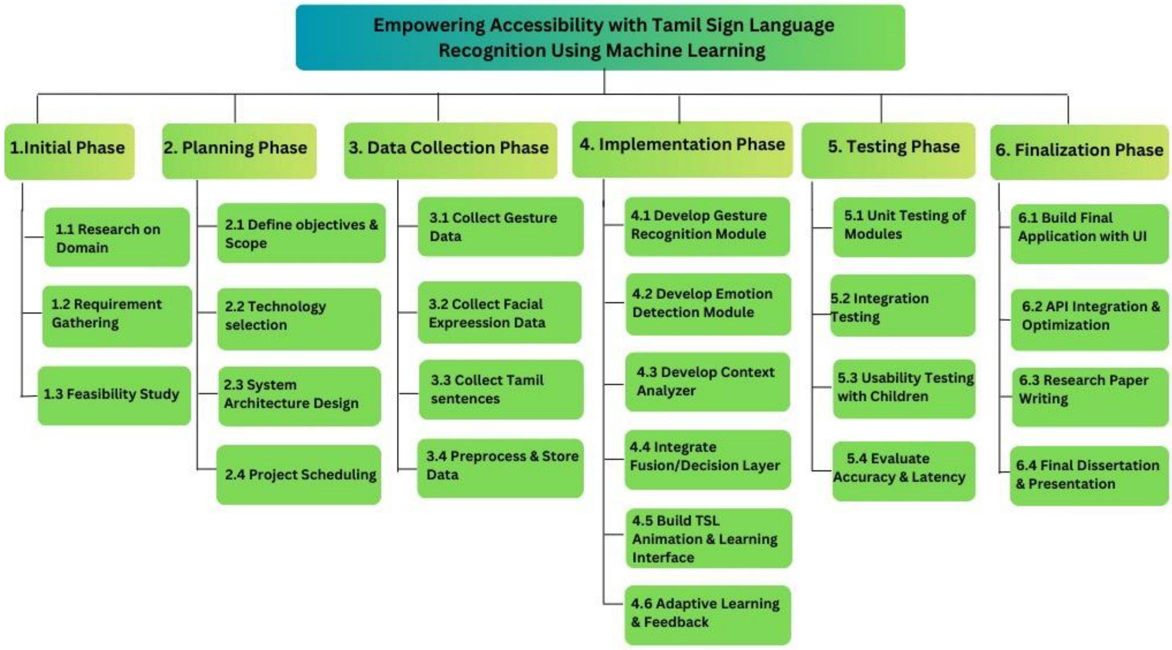

Figure 3.3 Work Breakdown Structure

3.9 Gantt Chart

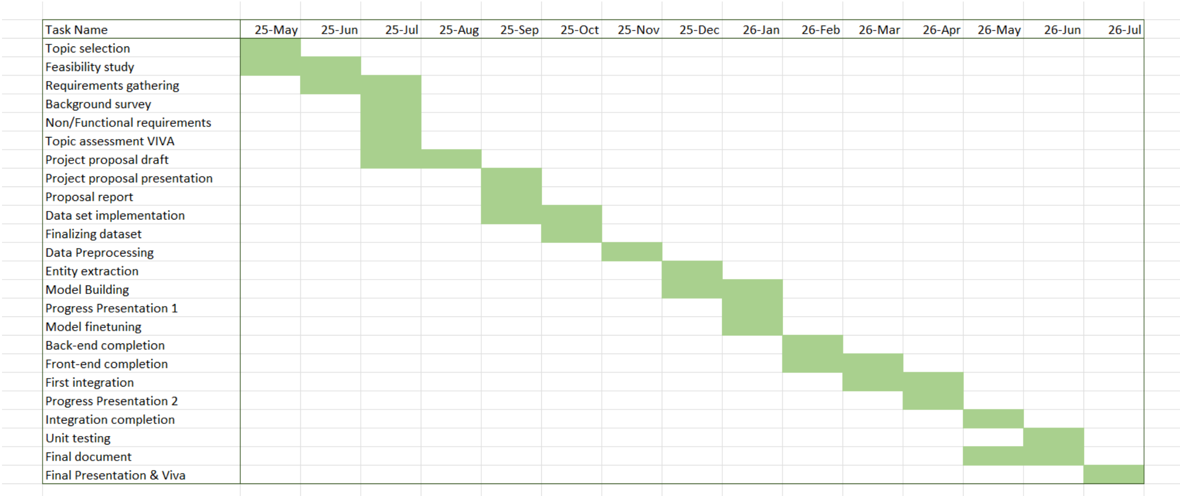

Figure 3.4 Gantt Chart

3.10 Project Requirement

3.10.1 Functional Requirements

- Provide structured TSL lessons (static ang dynamic signs).
- Support real-time practice with corrective feedback.
- Enable conversational practice with Tamil text/speech ↔ TSL.
- Adapt lesson sequencing and activity type to learner progress.
- Detect engagement and adjust activities accordingly.
- Provide teacher/parent dashboards with actionable insights.

3.10.2 Non-Functional Requirements

- Performance**:** The system should be very fast. When a learner signs, feedback should come almost immediately (within a fraction of a second). It should also be accurate in recognizing signs.

- Scalability**:** Many learners from different places should be able to use the platform at the same time without the system slowing down.

- Accessibility**:** The platform should be easy to use for everyone. It should follow international accessibility standards.

- Security**:** All data shared between users and the system should be safe. This means using secure connections (HTTPS), keeping information encrypted, and protecting learner privacy.

- Reliability**:** The system should always be available (almost no downtime). Learners should also be able to continue lessons offline if internet is not available.

4 DESCRIPTION OF PERSONAL AND FACILITIES

Registration No	Name	Task Description
IT22546166	Nethmi G.K.	• System Design & Development: Design and implement the TSL Learning & Education Platform as a PWA with Tamil-first user interface. • Lesson & Task Creation: Build the step-by-step learning track with motion-rich demos and interactive screen-based tasks. • Personalization & Recovery: Implement BKT with spaced repetition to deliver weekly personalized lesson packs, and add adaptive recovery patterns when learners fall off track. • Mentor Integration: Develop Mentor Chat for teachers to send micro-lessons and tips, and connect alerts from the insights service to mentors. • Progress Tracking: Provide checkpoints, badges, and dashboards to track learner progress and highlight weak areas. • Offline & Accessibility: Ensure offline packs for low-bandwidth users, Tamil-first localization, and simple, accessible design for all learners. • Evaluation & Testing: Conduct user testing with educators and hearing-impaired learners, measure learning gain, usability (SUS), and system performance, and refine based on feedback.

Table 4.1 Description of personal and facilities

5 BUDGET AND BUDGET JUSTIFICATION

The total estimated budget for the Tamil Sign Language (TSL) Audio-to-Sign Conversion System is LKR 15,500, carefully distributed across four main categories. This allocation ensures the successful collection of data, development of the system, testing, and deployment in real-world contexts.

1. Data Collection - LKR 4,000

The system requires high-quality Tamil speech data and corresponding Tamil Sign Language gesture datasets for training and validation. Since TSL resources are limited, part of this budget will be used for curating datasets through collaboration with educators, deaf schools, and community experts. Additional costs may include transcription, annotation, and basic preprocessing of audio and gesture recordings.

2. Software Development - LKR 4,500

The development stage includes:

- Implementing the speech recognition module (local ML-based ASR trained on Tamil datasets).
- Training logistic regression models for mapping tokenized Tamil text to TSL gestures.
- Developing a user-friendly API to connect the backend modules with the frontend interface.
- Designing the frontend application using React.js to visualize real-time gesture outputs.
- Backend integration through frameworks like Spring Boot with MySQL for database management.

This budget ensures the availability of development tools, libraries, and potential costs for external APIs during prototyping.

3. Testing and Validation - LKR 4,500

Testing is crucial to evaluate system accuracy, performance, and usability. This includes:

- Accuracy checks on speech-to-text recognition under different accents and noisy conditions.
- Validation of gesture prediction outputs for linguistic correctness.
- User acceptance testing with members of the Tamil-speaking hearing-impaired community.
- Iterative improvements based on feedback from educators and accessibility experts.

This allocation also covers logistical expenses for user trials and validation sessions.

4. System Deployment – LKR 2,500

Deployment involves hosting the system on a secure and scalable platform. Hosting may include:

- Cloud deployment with API integration for easy accessibility.
- Providing access to schools, NGOs, and public organizations supporting hearing-impaired communities.
- Basic maintenance and monitoring of the deployed system to ensure reliability.

The allocation ensures that the system is accessible to its target audience in real-world settings.

This budget allocation of LKR 15,500 strikes a balance between research rigor and resource efficiency. By prioritizing lightweight models and scalable software design, the system remains affordable while maintaining academic and practical value. The distribution emphasizes data quality, robust software development, extensive testing, and accessible deployment, ensuring that the Tamil Audio-to-TSL system is not only technically sound but also socially impactful.

Category	Budget (LKR)
Data Collection	4,000.00
Software Development	4,500.00
Testing and Validation	4,500.00
System Deployment	2,500.00
Total	15,500.00

Table 5.1 Estimated budget table.

6 APPENDICES

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